TACT SERVICES PLC | TRADE INFRASTRUCTURE & VALUE CHAIN ADVISORY

TECHNICAL COMPLIANCE MANUAL: MULTI-COOPERATIVE & SME EXPORT READINESS DIAGNOSTIC MATRIX

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Programmatic Framework:	Ethiopia Export Development Program (EEDP) [Target Authority: TradeMark Africa (TMA)]
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Focus Core Parameters:	Ethio-Djibouti Trade Corridor & Berbera Sub-Hubs | Horticulture, Pulses, and Oilseeds

1. EXECUTIVE MANDATE & PROGRAMMATIC STRATEGY

This technical manual operationalizes the proprietary diagnostic methodology engineered by TACT Services PLC. It is specifically calibrated to fulfill TradeMark Africa's (TMA) core mandate: accelerating intra-African trade, expanding Ethiopia's cross-border export volumes, and embedding pro-poor, gender-inclusive, and climate-resilient sustainability loops directly into national trade infrastructures.

Under the rigorous evaluation parameters of the Ethiopia Export Development Program, this diagnostic framework provides an automated, empirically validated methodology to audit, benchmark, and transition 150+ corporate Small and Medium Enterprises (SMEs) and agrarian cooperative unions from baseline operating inefficiencies into high-tier, export-ready enterprises. By mapping micro-level processing capacities against macro-level corridor logistical constraints, the tool ensures that Ethiopian agro-processed goods safely overcome Technical Barriers to Trade (TBT) and fully comply with European Union (EU), United Kingdom (UK), and GCC market access standards.

2. DETAILED EXPORT DIAGNOSTIC ARCHITECTURE (THE 4-PILLAR MATRIX)

The framework evaluates export-seeking entities across four heavily weighted technical pillars. Each pillar breaks down into verifiable, field-level audit vectors that are quantified through automated, mobile-based data collection forms to eliminate subjective grading bias.

PILLAR A: SANITARY & PHYTOSANITARY (SPS) SYSTEMS, TBT, & QUALITY INFRASTRUCTURE (SQI)

- **Audit Vector A.1: Maximum Residue Limits (MRLs) & Farm-Level Chemical Controls** — Rigorous digital inspection of smallholder pesticide logs, agro-chemical storage protocols, and field application intervals. This vector guarantees that high-value horticulture, pulses, and oilseeds systematically fall below the strict MRLs enforced by the EU and international food safety authorities.

- **Audit Vector A.2: Farm-to-Fork Digital Traceability Infrastructure** — Verification of active batch-coding systems capable of routing a specific shipment from individual smallholder cooperative primary collection centers up to processing plants and final shipping container sealing. This eliminates cross-contamination risks.
- **Audit Vector A.3: Post-Harvest Processing, Storage & Cold-Chain Integrity** — Granular audits of packhouse sanitation, ambient moisture controls (critical for oilseeds and pulses to prevent aflatoxin development), and automated temperature tracking systems.
- **Target Certification Standards Alignment:** GlobalG.A.P, HACCP Principles, ISO 9001, and ISO 14001 compliance frameworks.

PILLAR B: VALUE CHAIN LOGISTICS, COST STRUCTURES, & CORRIDOR EFFICIENCY

- **Audit Vector B.1: First-Mile Aggregation Cost Optimization** — Quantifying financial friction points, farm-gate collection deficits, and local transit delays experienced during the aggregation of raw crops from rural smallholder plots to processing facilities.
- **Audit Vector B.2: Non-Tariff Barrier (NTB) & Customs Procedural Drag** — Empirical time-motion modeling of structural delays related to export documentation, customs clearance time loops, transit checks, and phytosanitary certificate issuance speeds along the Ethio-Djibouti Trade Corridor and the emerging Berbera Corridor Sub-Hubs.
- **Audit Vector B.3: Cold-Chain Corridor Logistics & Spoilage Modeling** — Cost-vs-time optimization algorithms mapping refrigerated and dry shipping container transit to substantially lower post-harvest spoilage rates before port arrival, preserving product margin.

PILLAR C: AGRO-INDUSTRIAL PROCESSING CAPACITY & VALUE ADDITION

- **Audit Vector C.1: Mechanical Sorting, Grading, and Market Specification Matching** — Evaluating automated sorting lines to ensure agricultural exports (pulses and oilseeds) precisely meet international buyer standards for purity, size grading, color consistency, and moisture content.
- **Audit Vector C.2: Floor Efficiency, Throughput Capacity, and Waste Mitigation** — Isolating and neutralizing processing-floor structural waste to lower unit costs, maximizing net profit margin retention for cooperative smallholder suppliers.
- **Audit Vector C.3: Industrial Quality Management & Laboratory Accreditation Systems** — Verification of processing facility compliance with Good Manufacturing Practices (GMP) and alignment with national and international laboratory testing accreditation bodies.

PILLAR D: INSTITUTIONAL GOVERNANCE, SOCIAL INCLUSION, & PUBLIC-PRIVATE DIALOGUE (PPD)

- **Audit Vector D.1: Pro-Poor Cooperative Governance & Revenue Distribution** — Assessing profit-sharing mechanisms within agrarian unions to confirm that premium export revenues directly reach and uplift smallholder farming families.
- **Audit Vector D.2: Gender Equity & Women's Economic Inclusion Safeguards** — Auditing institutional frameworks to ensure women processing operators and smallholder farmers

maintain equal pay, leadership representation, secure working conditions, and targeted technical capacity building.

- **Audit Vector D.3: Public-Private Dialogue (PPD) Data Readiness** — Evaluating the entity's structural capacity to collect and feed local trade-friction data (e.g., customs delays or unfair tax hurdles) directly into national-level policy reform platforms.

3. SCORING METHODOLOGY & RISK CLASSIFICATION GRID

Field data is funneled into an automated relational algorithm that computes a weighted Export Readiness Index (ERI) Score from 0% to 100%.

Score Band & Status	Operational Metric / Status	Strategic Action Loop
[80% – 100%] EXPORT READY (Green Band)	Full structural compliance with international SPS/TBT and corridor logistics standards.	Immediate integration into TradeMark Africa's international buyer matching networks, trade facilitation channels, and high-value export financing tracks.
[50% – 79%] CONDITIONAL (Yellow Band)	Viable production capacity, but compromised by isolated gaps (e.g., cold-chain deficits or excessive customs drag).	Channeled into specialized TACT Services capacity-building pipelines. Targeted interventions are deployed to upgrade quality systems and lower compliance costs.
[UNDER 50%] CRITICAL RISK (Red Band)	Structural vulnerability, extreme post-harvest waste, chemical non-compliance, or severe governance deficiencies.	Excluded from immediate export consideration. Diverted to fundamental value-chain development pipelines to rebuild primary agricultural production capability.

4. ADAPTIVE MEL WORKSPACE & DIGITAL DATA COUPLING

To fulfill TradeMark Africa's requirements for transparent contract performance and strict quality assurance, the diagnostic outputs are structured into an active relational database. Individual SME and cooperative metrics are aggregated into live interactive data pipelines using advanced Star-Schema Models.

This data strategy transforms raw field audits into macro-level policy assets. It allows TMA, national counterparts, and international donors to trace exact capacity progress, identify systemic trade bottlenecks along the Ethio-Djibouti corridor in real-time, and make fast, evidence-based, adaptive management adjustments to safeguard program impact.